

# Adem Toumi

## Artificial Intelligence & Data Science Engineer

+213 795 016 471 | Biskra, Algeria | a.toumi@esi-sba.dz | LinkedIn | GitHub

### PROFESSIONAL SUMMARY

AI & Data Science Engineer covering the full spectrum of modern artificial intelligence — deep learning, large language models, computer vision, natural language processing, data engineering, and software development. Graduating July 2026 from ESI-SBA (State Engineer's Degree & M.Sc., Artificial Intelligence & Data Science).

### EDUCATION

#### Higher School of Computer Science (ESI-SBA)

State Engineer's Degree & M.Sc. in Artificial Intelligence and Data Science

Sidi Bel Abbès, Algeria

Sep. 2021 – Jul. 2026

- **Dual Thesis (Engineering & M.Sc.):** *Log-Based Anomaly Detection in Distributed Systems* — research and engineering work submitted for both the State Engineer's Degree and the Master's degree; benchmarked ML, deep learning, and transformer-based approaches across industrial-scale log datasets and built a real-time streaming pipeline with cloud inference and an explainability dashboard.
- **Coursework:** Machine Learning · Deep Learning · NLP · Computer Vision · Distributed Systems · Big Data Engineering · Software Engineering.

### TECHNICAL SKILLS

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|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming Languages:</b>      | Python · TypeScript · JavaScript · Dart · Java · SQL                                                                          |
| <b>Machine Learning:</b>           | PyTorch · TensorFlow · Scikit-learn · XGBoost · Supervised & Unsupervised Learning · Model Evaluation                         |
| <b>Deep Learning &amp; Vision:</b> | Computer Vision · Object Detection & Tracking · Deep Learning · OpenCV                                                        |
| <b>NLP &amp; Generative AI:</b>    | Natural Language Processing · Large Language Models · Retrieval-Augmented Generation · Semantic Search · Explainable AI (XAI) |
| <b>Data Engineering:</b>           | Apache Kafka · Apache Spark · PySpark · Hadoop · Apache Airflow · Data Pipelines                                              |
| <b>Databases &amp; Storage:</b>    | PostgreSQL · Firebase · Relational & Non-Relational Databases                                                                 |
| <b>Software &amp; Deployment:</b>  | FastAPI · Node.js · React · Next.js · Flutter · Docker · Git · Linux · MLflow                                                 |

### RESEARCH & INDUSTRY EXPERIENCE

#### Research Intern – Computer Vision & Embedded Systems

Feb. 2025 – Jun. 2025

Electronic Systems Research & Development Unit (URD Laboratory), Sidi Bel Abbès, Algeria

- Developed a real-time aerial surveillance system for UAVs — trained object detection models on aerial datasets (mAP@0.5  $\approx$  0.85), integrated dual tracking pipelines, and deployed the full inference stack on edge hardware (Raspberry Pi 5 + Intel Movidius NCS 2).
- Designed the complete hardware-software architecture: custom UDP video streaming to a Ground Control Station, automated gimbal control that keeps tracked targets centered, and an interactive operator interface with live telemetry.

#### IT Intern – Database & Network Administration

Sep. 2024

Sonatrach, Direction Maintenance, Biskra, Algeria

- Optimized SQL queries, maintained database schemas, and automated backup scripts in an industrial environment.
- Supported network administration and system monitoring tasks.

### SELECTED PROJECTS

#### Artificial Intelligence & Data Science

##### Log-Based Anomaly Detection in Distributed Systems

Python, PyTorch, Kafka, PostgreSQL, Streamlit

Engineering & Master's Graduation Thesis

- Engineered a complete real-time anomaly detection pipeline for distributed infrastructure: log streams flow through

Apache Kafka, ML and transformer-based models run cloud inference in parallel, results persist to a database, and a live dashboard surfaces alerts with per-model metrics.

- Integrated XAI (Explainable AI) using SHAP and LIME — token-level attributions are computed on demand and displayed inline in the dashboard, making every anomaly flag fully interpretable.

**KnowledgeHub AI – Enterprise RAG Platform**

*Python, FastAPI, React, LLMs, Docker*

- Built a production-ready document intelligence platform: organizations upload their knowledge base, ask questions in natural language, and receive grounded answers with exact source citations.
- Delivered the complete stack — hybrid search engine, FastAPI backend, React frontend, Docker Compose deployment, and a comprehensive automated test suite.

**Object Detection & Tracking for UAV Applications**

*Python, PyTorch, Computer Vision, Edge AI*

- End-to-end aerial surveillance system covering object detection, multi-target tracking, edge deployment, and a Ground Control Station interface — full detail in Research Experience above.

**Malicious URL Detection – Security Classifier**

*Python, Scikit-learn, XGBoost, TensorFlow*

- Built a multi-class URL classifier to detect phishing, malware, and defacement threats; engineered a hybrid feature set combining text representations with structural URL descriptors — best model reached 96.8% accuracy.

**Paraphrase Detection – NLP Similarity System**

*Python, PyTorch, NLP*

- Built a sentence-level semantic similarity system using a Siamese deep learning architecture with attention mechanisms, trained on large-scale NLP benchmarks.

**Software Engineering**

**HajjPortal – Trilingual Management Platform**

*TypeScript, Next.js, React, Socket.io*

- Built a full-stack web platform for pilgrimage management — registration, reservations, lottery, and journey tracking — across a pilgrim-facing application and a separate admin dashboard.
- Implemented trilingual support (Arabic, English, French) with dynamic RTL/LTR switching, JWT authentication, and real-time updates via WebSockets.

**KidLink – School Communication Platform**

*Dart, Flutter, Firebase*

- Built a cross-platform mobile app connecting parents, teachers, and administrators in primary schools — role-based interfaces, real-time push notifications, and a cloud backend.

**LANGUAGES**

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Arabic (Native)

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French (Fluent)

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English (Professional)